

17/10/22

Roll No.

TME-501

B. TECH. (ME/AME/Robotics) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

HEAT & MASS TRANSFER

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define modes of heat transfer conduction, convection and radiation (CCR) and their governing equations. (CO1)

OR

- (b) Two slabs are placed in contact, but due to roughness, only 40% of area is in contact and the gap in the remaining area is 0.2 mm thick and is filled with air. The slabs are 10 cm thick each and their conductivities are 15.5 W/mK and 200 W/mK. The temperature of the face of the hot surface is at 250 °C and the outside surface of the other slab is at 35°C. Determine the heat flow and the contact resistance. The conductivity of air is 0.0321 W/mK. (CO1)

P. T. O.

2. (a) What is Critical thickness of insulation ? What is the advantages of critical radius of insulation ? (CO1)

OR

- (b) A copper pipe carrying refrigerant at -25°C is 12 mm in OD and is exposed to convection at $55\text{ W/m}^2\text{K}$ to the air at 30°C . It is proposed to apply insulation of conductivity 0.5 W/mK . Determine the thickness beyond which the heat gain will be reduced. Calculate the heat gain for 2.5 mm, 5.0 mm and 7.5 mm thickness for 1 m length. The convection coefficient remains constant. (CO1)

3. (a) What is fin ? Derive an expression for temperature profile for an infinitely long fin. (CO2)

OR

- (b) A long rod 15 mm square section made of low carbon protrudes into air at 30°C from furnace wall at 250°C . The convective heat transfer coefficient is estimated at $20\text{ W/m}^2\text{K}$. The conductivity of the material is 55.9 W/mK . Determine the location from the wall at which the temperature will be 80°C . Calculate the temperature at 90 mm from the base. (CO2)

4. (a) Define and the formula for the following terms : (CO1, CO2)
- (i) Reynolds number
 - (ii) Nusselt number
 - (iii) Biot number
 - (iv) Fourier's number

(3)

OR

- (b) Derive the general three dimension heat conduction equation in Cartesian coordinates system and also discuss various cases.

(CO1, CO2)

5. (a) What is thermal conductivity of material ? Write the unit of thermal conductivity and the value of thermal conductivity of diamond, copper, mild steel and aluminum.

(CO1, CO2)

OR

- (b) What do you mean by transient heat conduction ? How is rate of heat transfer calculated using Lumped parameter analysis ?

(CO1, CO2)